

Abhinav Singh

🔗 [Hazard787](#) 📧 [solankiabhinav](#) ✉ [abhinav787agwanpur@gmail.com](#) ☎ [+91 9693123026](#)



Education

Bachelor of Technology in Computer Science & Engineering
Ramaiah Institute of Technology, Bangalore
Class XII (CBSE) — MS Memorial Academy, Patna, Bihar
Class X (CBSE) — St. Michaels High School, Patna, Bihar

CGPA: 8.41/10
2023 - 2027
81.50%
2022
87.83%
2020

Technical Skills

Programming Languages: Python, Java, C, JavaScript, SQL
AI & Computer Vision: YOLOv8/v11, OpenCV, Pose Estimation, PyTorch, MediaPipe, CUDA
Libraries & Frameworks: Spring Boot, React, Node.js, Express.js, Socket.IO, FastAPI, Flask, scikit-learn, Pandas, NumPy
AI & Data Technologies: RAG, Sentence Transformers, ChromaDB, LLMs, NLP, Matplotlib, Seaborn
Tools & Platforms: Git/GitHub, VS Code, Jupyter Notebook, Vercel, Railway, Power BI, Excel
Core Competencies: Data Structures & Algorithms, Machine Learning, Web Development, Data Analysis

Projects

- AI-Based Classroom Behavior Monitoring & Optimized Storage** Feb 2026 - May 2026
Computer Vision & AI Inference Project
- Developed a real-time monitoring system using **YOLOv8-Pose** to identify behavioral anomalies across 35-bench grid layouts.
 - Implemented **Hybrid Logic** utilizing skeletal aspect ratios and perspective-calibrated ROI polygons to reduce false positives by 40%.
 - Engineered an event-driven recording module that optimizes media storage, reducing redundant data retention by over **90%**.
 - Technologies:** Python, YOLOv8, OpenCV, NumPy, JSON, Git.
- ScholarSync - RAG-Based Academic Writing Assistant** Jun 2026 - Jul 2026
Generative AI & Retrieval-Augmented Generation
- Built a **RAG-based academic assistant** that ingests research papers, performs semantic retrieval, and generates citation-grounded answers and literature reviews.
 - Implemented a document pipeline using **PyMuPDF, Sentence Transformers, and ChromaDB** for PDF parsing, recursive chunking, local embeddings, and persistent vector storage.
 - Developed **FastAPI** endpoints for document ingestion, grounded Q&A, summarization, comparison, literature review, and citation generation using the **Groq Llama 3.3 70B API**.
 - Built a **Streamlit interface** supporting multi-paper uploads and source-aware responses with paper and page-level citations.
 - Technologies:** Python, FastAPI, Streamlit, ChromaDB, Sentence Transformers, PyMuPDF, Groq API, Llama 3.3 70B.
- Real-Time Multiplayer Card Game** Jul 2026 - Aug 2026
Full-Stack & Real-Time Web Application
- Built and deployed a real-time multiplayer card game supporting **4 concurrent players** with room-based matchmaking and live game synchronization.
 - Developed a **server-authoritative game engine** handling room management, bidding, turn progression, card validation, scoring, and multi-round gameplay with **server-side move validation**.
 - Implemented **Socket.IO event-driven communication** to synchronize game state, player actions, and private player hands across connected clients in real time.
 - Technologies:** React, Vite, JavaScript, Node.js, Express.js, Socket.IO, Vercel, Railway.

Additional Information

- Languages:** English (Fluent), Hindi (Native)
- Interests:** Gaming, Cooking, Table Tennis, Football